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Lecture 01A

What can be a Vector?

MH1201 Linear Algebra 2

<https://www.wooclap.com/>

HIDDEN

Lecture 01 – Vector Spaces+Subspaces

- Course Admin
- Vector Addition and Scalar Multiplication
- Examples of Vector Space
 - $\mathcal{M}_{m \times n}(\mathbb{F})$: $m \times n$ -Matrices
 - $\mathcal{F}(X, \mathbb{F})$: Functions from X to $\mathbb{F}$
- Axioms of Vector Space
- Properties of Vector Space
- Subspace

Course Admin

Instructors

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- Teaching Assistants
 - Wilton Kim
 - John Ng
 - Joel Nathanael
 - Chew Yuan Xiang

Class Schedule (AY2023/2024 S2)

Lectures will be conducted both **in-person** and **"livestreamed" via Zoom.**



- <https://ntu-sg.zoom.us/j/91450438196> (Passcode: basis)
- **Mon** – 1430hrs to 1520hrs
- **Thu** – 1430hrs to 1620hrs

Tutorials (Tue/Wed) conducted **in-person** ~~at CC-Collab1~~

Course Management (AY2023/2024 S2)



- **Google Colab Notebooks**

- https://colab.research.google.com/drive/1C_JU8ZDFwZ9M5_gghdFy_NA4uLybVktM?usp=sharing
- Lecture Notes – Blank, Handwritten
- Tutorials – PDF, Blank Notes, PDF Solutions, Video Solutions

- **NTULearn**

- Not many links.
- **IMPORTANT:** Wooclap links. **Login via NTULearn!**

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Assessment

- Finals (50%)
 - CLOSED book (no cheatsheet)
- Midterm (30%)
 - CLOSED book (no cheatsheet)
 - Cover vector space, vector subspace, span, linear independence, bases, dimension, linear maps
 - 29 Feb 2024 (Thu), 1500hrs to 1600hrs
 - Hall C
- Wooclap (20%)

Wooclap Sessions

The Wooclap logo, featuring the word "wooclap" in white lowercase letters on a blue rectangular background.

- We will use wooclap for every lecture
- Simple short questions >> immediate feedback
- Grading System
 - If you answer N questions correctly throughout the semester, your score will be

$$\min \left\{ \frac{N}{4}, 20 \right\}$$

- Wooclap session: 20% of final grade.
- There are **NO** makeup sessions.

Telegram Group



- Join the group to communicate with me and tutors during lectures or tutorials or at other times.
- I will use this platform and NTULearn to make announcements.
- <https://t.me/+veyL820nwPliMml1>



What can be a Vector?

Examples and Axioms

What Else can be a Vector?

Linear Algebra 1

- $\mathcal{V} = \mathbb{R}^n \rightarrow n\text{-tuple}$
- $\oplus$ is component-wise vector addition
- $\otimes$ is component-wise scalar multiplication.

$$\vec{u} = (u_1, u_2, u_3 \dots u_n)$$

$$\vec{v} = (v_1, v_2, v_3 \dots v_n)$$

$$\vec{u} \oplus \vec{v} = (u_1 + v_1, u_2 + v_2 \dots u_n + v_n)$$

$$\lambda \otimes \vec{u} = (\lambda u_1, \lambda u_2, \dots \lambda u_n)$$

$\hookrightarrow$ a real number

Linear Algebra 2 (THIS COURSE!)

- $\mathcal{V} \rightarrow$ can be "anything"

- $\oplus$
 - $\otimes$
- } Task: Define these two operations for "anything"

Vectors in this course:

$\rightarrow$ matrices

$\hookrightarrow$ symmetric matrices...

$\rightarrow$ functions

$\hookrightarrow$ polynomials

Today:

Define $\oplus, \otimes$ for
matrices and functions

$$(\mathcal{M}_{m \times n}(\mathbb{F}), \oplus, \otimes)$$

$\mathcal{M}_{m \times n}(\mathbb{F})$ is set of matrices with dim $m \times n$ whose entries belong to $\mathbb{F}$.
 (Annotations: curly M, # row, # column)

$\rightarrow \mathbb{F}$ is a field (TA; PI) Too abstract. Please ignore

$\rightarrow$ In this course, $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$

hat $\leftarrow \hat{A} = \begin{bmatrix} a_{11} & a_{12} & \dots \\ a_{21} & a_{22} & \dots \\ \dots & \dots & \dots \end{bmatrix} = [a_{ij}]$

$$(\mathcal{M}_{m \times n}(\mathbb{F}), \oplus, \otimes)$$

$\mathcal{M}_{m \times n}(\mathbb{F})$ is the set of $m \times n$ matrices whose entries belong to $\mathbb{F}$.

$\oplus$: vector addition is defined as follow:

$$\overset{m \times n}{\hat{A}} \oplus \overset{m \times n}{\hat{B}} \triangleq \overset{m \times n}{\hat{C}}$$

$$[a_{ij}] \oplus [b_{ij}] \triangleq [a_{ij} + b_{ij}]$$

⚡ So here, $\mathcal{M}_{m \times n}(\mathbb{F})$ is **closed under $\oplus$**
because adding two $m \times n$ matrix gives a $m \times n$ matrix.

$$(\mathcal{M}_{m \times n}(\mathbb{F}), \oplus, \otimes)$$

$\mathcal{M}_{m \times n}(\mathbb{F})$ is the set of $m \times n$ matrices whose entries belong to $\mathbb{F}$.

$\otimes$: Scalar Multiplication for matrices is defined as :

$$\lambda \otimes \hat{A}$$

↙ scalar
→ vector (in this case, this is a matrix)

$$\lambda \otimes [a_{ij}] = [\lambda a_{ij}]$$

Here, $\mathcal{M}_{m \times n}(\mathbb{F})$ is closed under scalar multiplication because multiplying $m \times n$ matrix by a number λ yields an $m \times n$ matrix.

$$(\mathcal{M}_{m \times n}(\mathbb{F}), \oplus, \otimes)$$

$\mathcal{M}_{m \times n}(\mathbb{F})$ is the set of $m \times n$ matrices whose entries belong to $\mathbb{F}$.

* Besides $\mathcal{M}_{m \times n}$ is closed $\oplus$ & $\otimes$

we have to check many other axioms

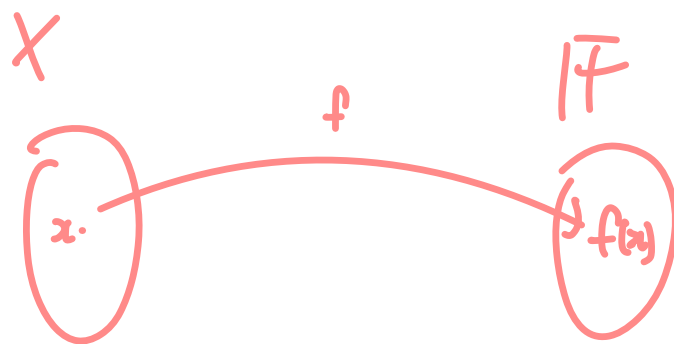
TBEL, To be explained later

$$(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$$

→ curly F

$\mathcal{F}(X, \mathbb{F}) \triangleq \{\text{all functions whose domain is } X \text{ and codomain is } \mathbb{F}\}$

if $f \in \mathcal{F}(X, \mathbb{F})$,



$$(\mathcal{F}(\mathbb{R}, \mathbb{R}), \oplus, \otimes)$$

domain codomain

$\mathcal{F}(X, \mathbb{F})$ is the set of functions that map elements in X to elements in $\mathbb{F}$.

Examples of functions / vectors in $\mathcal{F}(\mathbb{R}, \mathbb{R})$

- $f(x) = 1 + 2x + x^2$
- $g(x) = 1 - 2x + \frac{1}{2}x^2$
- $F_1(x) = \sin x, F_2(x) = \sin 2x, \dots$
- $G_1(x) = \cos x, G_2(x) = \cos 2x, \dots$

$$(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$$

Qn: How to add functions?

Define $f \oplus g$ to be the function

$$(f \oplus g)(x) = f(x) + g(x)$$

Example:

$$f(x) = 1 + 2x + x^2$$

$$g(x) = (-2x + \frac{1}{2}x^2)$$

$$f \oplus g(x) = 2 + \frac{3}{2}x^2$$

$\mathcal{F}(X, \mathbb{F})$ is the set of functions that map elements in X to elements in $\mathbb{F}$.

Qn: Does $f \oplus g$ belong to $\mathcal{F}(X, \mathbb{F})$?

Yes. Because $f(x) + g(x) \in \mathbb{F}$.

$\Rightarrow f \oplus g$ is closed under $\oplus$.

$$(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$$

$\mathcal{F}(X, \mathbb{F})$ is the set of functions that map elements in X to elements in $\mathbb{F}$.

Qn: How to define $\otimes$?

Define $\lambda \otimes f$
 number $\downarrow$ function

$$(\lambda \otimes f)(x) = \underbrace{\lambda f(x)}_{\text{multiplication of numbers}}$$

Example:

$$F_1(x) = \sin x$$

$$2F_1(x) = 2\sin x$$

Since $\lambda f(x) \in \mathbb{F}$, we have $\lambda \otimes f \in \mathcal{F}(X, \mathbb{F})$,

$\therefore \mathcal{F}(X, \mathbb{F})$ is closed under $\otimes$.

$$(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$$

$\mathcal{F}(X, \mathbb{F})$ is the set of functions that map elements in X to elements in $\mathbb{F}$.

- We have defined vector addition $\oplus$ and scalar multiplication $\otimes$ for $\mathcal{F}(X, \mathbb{F})$ – the set of all functions whose domain is X and codomain $\mathbb{F}$.
- To say that $(\mathcal{F}(X, \mathbb{F}), \oplus, \otimes)$, we need to check the ten axioms